Tiny Legs or Tiny Wheels? Comparing locomotion morphologies for 100 mg quadrupeds using real-to-sim system identification

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Abstract*— Designing effective locomotion for gram and sub-gram terrestrial robots is challenging due to lower inertias, contact-dominated losses, higher step frequencies, fabrication tolerances, and limited onboard resources. This paper presents a comparative analysis of the Milliquad family of $\approx 100\,\mathrm{mg}$ magnetically-actuated quadrupeds with leg morphologies spanning the leg-wheel spectrum, from a single-spoke appendage to a continuous wheel. To move beyond empirical comparisons, we also develop a system identification approach towards an effective MuJoCo model with uncertain contact and magnetic parameters optimized to fit experimental results. The resulting simulation is then used as an analysis tool to estimate mechanical cost of transport (COT), which is difficult to measure directly in untethered robots at this scale. The simulations qualitatively capture robot locomotion behavior and quantitatively agree with many of the experimental result trends. Experiments and simulations show that wheeled morphologies are still fast and efficient at these small scales over flat terrain ($v = 449\,\mathrm{mm\,s^{-1}}$, COT $= 0.45$ at a drive frequency of $30\,\mathrm{Hz}$) but are far less reliable on more complex terrain, including $1\,\mathrm{mm}$ steps. The double-spoke morphology provides the strongest overall terrain robustness, including the fastest step traversal and an $80\,\%$ success rate on rough terrain with highly repeatable speed. Source code used for data processing and simulation will be available upon publication.

I. Introduction

Small-scale terrestrial robots can access confined and cluttered environments that are difficult for larger platforms, enabling applications such as inspection, monitoring, and search through complex environments like rubble or industrial infrastructure. At these scales, limited on-board power, actuation, and sensing constrain the mechanisms and control strategies that are practical, making the locomotion morphology (e.g., legs or wheels) an important design problem. For larger ground robots, the tradeoffs between wheels and legs are well established. Wheeled platforms are typically fast and energy efficient on smooth ground [1–5], while legs can improve mobility on irregular terrain by selectively making and breaking contact [1,6,7]. Hybrid wheeled-legged systems combine rolling efficiency with the ability to navigate rougher terrain [1].

It is less clear whether these established tradeoffs translate to gram and sub-gram terrestrial robots [8]. As robots become smaller, fabrication tolerances become increasingly consequential [9], sensing and actuation pipelines for precise control are often minimal or absent [10], and contact-dominated effects such as friction and damping begin to dominate system losses and robot dynamics [11–13]. In fact, legged designs with flexure joints have typically been preferred over wheeled designs at these small scales, in part because rotary bearings are expected to suffer from stiction and excessive friction as surface forces increase relative to inertial forces [8,14]. In addition, terrain features that are negligible at larger scales can become comparable to leg length at smaller scales, so significant "roughness" and steps exist in more environments [15]. These scale effects motivate a systematic investigation of wheel-leg morphology choices at smaller sizes, along with an analysis approach that can go beyond empirical results and help explain why certain morphologies succeed or fail.

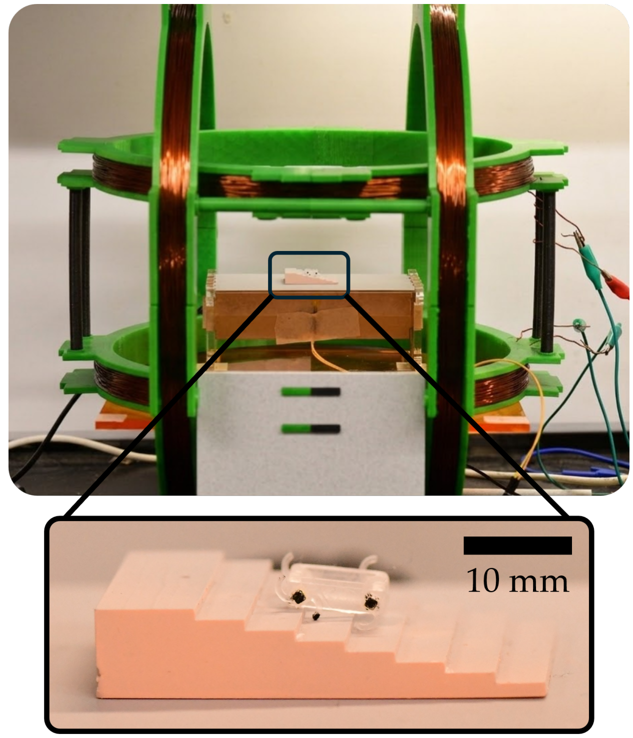

Fig. 1. A magnetically-actuated $100\,\mathrm{mg}$ Milliquad robot with double-spoke leg configurations traversing steps in the Helmholtz coil setup. The robot's hip and leg were colored for tracking purposes.

In this work, we quantify the locomotion characteristics of the Milliquad family of magnetically actuated quadrupeds with varied leg morphologies, ranging from a single-spoke leg to a continuous wheel (Fig. 2). The robots have magnets embedded into their hip joints and are driven using a rotating magnetic field provided by two pairs of Helmholtz coils. This setup allows us to study the robot's locomotion dynamics without the need to include onboard power and actuation systems. The Milliquad platform builds on prior magnetically actuated quadrupeds at similar and smaller sizes [12,16,17], but extends these efforts by providing a systematic morphol-

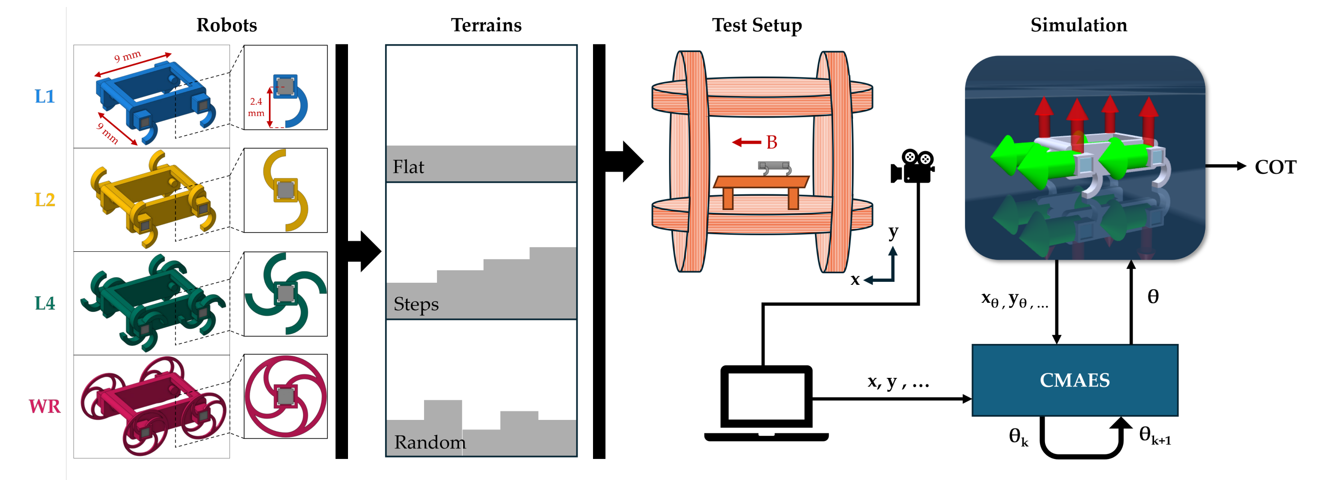

Fig. 2. Four robots with different leg-wheel morphologies were fabricated and tested on three terrains: flat ground, steps, and random rough terrain. The terrains were placed within a 2-DOF Helmholtz coil system for magnetic actuation. Each trial was recorded using a high-speed camera. The recorded videos were subsequently analyzed to extract kinematic data points for performance evaluation. The extracted data points were further used to identify parameters for a simulation model using the CMA-ES optimization algorithm.

ogy comparison over multiple terrains.

Another remaining challenge for design insight at this scale is measurement. Gathering data on energetic cost and contact forces would be useful in understanding morphology dependence on locomotion, but these data are challenging to obtain from robots at sub-gram scales. For magnetically actuated systems in particular, direct measurements of electrical power in the coils is not sufficient to determine the mechanical work delivered to the robot. Recent work using multi-body simulators like MuJoCo [18] have been used to study locomotion in compliant multi-legged robots [19], soft millirobots [10], and even fruit flies [20]. However, it is typically challenging to determine appropriate contact parameters in these simulations, especially for low-mass systems [20]. To address our measurement challenges, we fit the most uncertain parameters of a Milliquad MuJoCo model (contact and magnetic parameters) to experimental trajectory data using a global optimization algorithm. The resulting optimized simulation is then used as an analysis tool to estimate quantities that are challenging to measure directly, including mechanical energy expenditure.

Together, these efforts make two key contributions. First, we provide a comparative analysis of four $\approx 100\,\mathrm{mg}$ robots with different wheel-leg morphologies across flat ground, steps, and rough terrain, characterizing speed and stability trends across different drive frequencies. Second, we introduce a system-identification workflow that combines physics-based modeling with global optimization to align simulation with magnetically-driven small-scale robot experiments, enabling quantitative estimation of hard-to-measure quantities such as the mechanical cost of transport.

II. Methods

Our approach combines both hardware experiments and physics-based simulation (Fig. 2). We fabricate four Milliquad morphologies and actuate them using a custom 2-DOF Helmholtz coil system. Each trial is recorded with a high-speed camera and the applied magnetic field is logged. From the video, we extract robot kinematics, which serve as both performance metrics and as reference data for system identification. We then fit uncertain contact and magnetic parameters of a MuJoCo model using global optimization to replicate the robot's average velocity, and use the simulation to estimate quantities like the mechanical cost of transport (COT). The following subsections detail the design choices and specifications of both hardware and simulation.

A. The Milliquads

1) Robot morphologies: The Milliquad quadrupedal robots have a $9\,\mathrm{mm} \times 9\,\mathrm{mm}$ body with a $1.5\,\mathrm{mm}$ ground clearance and a mass of $108.4\,\mathrm{mg}$ for single-spoke leg (L1), double-spoke leg (L2), and quadruple-spoke leg (L4), and a mass of $105.9\,\mathrm{mg}$ for wheeled-robot (WR). The legs and wheels of all four robots were designed to have identical moments of inertia about their respective rotation axes. A $1\,\mathrm{mm}$ cube N50 Nickel-plated magnet (SuperMagnetMan) is embedded into each rotational hip joint (Fig. 2). Each robot is 3D-printed using a high-resolution resin printer (Pico 2 HD, Asiga) and assembled by hand. We evaluate four appendage geometries spanning the wheel-leg spectrum: L1, L2, L4, and WR. Each spoke is a semi-circle of diameter $2.4\,\mathrm{mm}$ attached to the hub at one end as illustrated in Fig. 2. The double and quadruple spoke variations have spokes spaced $180°$ and $90°$ apart, respectively. For WR, the wheel rim connects the four ends of the L4s, to create a wheel with a radius of $2.4\,\mathrm{mm}$.

Two sets of robots of each wheel-leg geometry were fabricated, totaling eight robots. After comparing robots with the same morphologies to determine that there were no significant differences due to fabrication, one set of robots was used for the remaining trials.

2) Magnetic actuation and coil drive: The magnetic torque exerted on the Milliquads' hip magnets by a uniform external magnetic field is modeled using the point-dipole model, similar to [12]:

$$\tau_{\mathbf{drive}} = \mathbf{m} \times \mathbf{B_{ext}} \tag{1}$$

where $\mathbf{m}$ is the magnetic dipole, and $\mathbf{B_{ext}}$ is the uniform external magnetic field. Under load, the magnet tracks the rotating field with a phase lag, resulting in a nonzero torque as defined in (1). The external magnetic fields are provided by two orthogonal Helmholtz coil pairs with diameters $30\,\mathrm{cm}$ and $36\,\mathrm{cm}$ (Fig. 1). Both coils are wrapped around 3D-printed rings: the larger coil holds 120 turns of 18 AWG wire with a total resistance of $7.1\,\Omega$, while the smaller coil has 120 turns with a total resistance of $6.4\,\Omega$. Each pair is separated by a distance equal to the coil radius to generate a uniform magnetic field within the center of the workspace. We verified the uniform-field region using a 3-axis magnetometer (TLE493d, Infineon), confirming that the robots can travel more than 10 body lengths without gradients that could introduce net translational forces.

The coils are driven through H-bridges (BTS7960) connected to a DC power supply (XPF 60-20DP, Sorensen) using PWM from a microcontroller (Arduino Uno REV3). Because field amplitude is frequency dependent, we calibrate each axis using the magnetometer and scale PWM commands so that the measured field amplitude matches the commanded field amplitude across frequencies. We command a rotating field in the x-y plane at frequency f_{drive} by first evaluating the desired field angle and using that to calculate the desired field with magnitude B_0 from each coil:

$$\theta_B(t) = 2\pi f_{drive} t \tag{2}$$
$$B_x(t) = B_0 \sin(\theta_B(t)) \tag{3}$$
$$B_y(t) = B_0 \cos(\theta_B(t)). \tag{4}$$

Using Biot-Savart and Ohm's laws, the PWM voltage is calculated as

$$V(t) = B(t) \left(\frac{5\sqrt{5}\rho r^2 \pi}{4\mu_0 A} \right) \tag{5}$$

where ρ is the resistivity of the wire, r is the radius of the coil, $\mu_0 = 4\pi \times 10^{-7}\,\mathrm{T\,m\,A^{-1}}$ is the vacuum permeability constant, and A is the cross-sectional area of the wire. A field amplitude of $2\,\mathrm{mT}$ was used for all experiments.

B. Experimental protocol and metrics

1) Terrains: We evaluate locomotion across three types of terrain: flat ground, steps, and random-square rough terrain. All surfaces are coated with a 1200-grit polymer surface treatment (Mr.Surfacer 1200, Mr.Hobby) to reduce friction variability across terrains. Steps are 3D-printed (Pico 2 HD, Asiga) and are parameterized by number of steps (8 steps), a step height of $1\,\mathrm{mm}$, and a step length of $4.5\,\mathrm{mm}$ (half of the robot body length). The step height and length were selected based on preliminary testing. This configuration was chosen because it induces qualitatively different locomotion behaviors across robot morphologies, thereby maximizing the observable performance differences. The rough terrain is a 3D-printed (Pico 2 HD, Asiga) field of square pillars with different heights. The rough terrain is parameterized by the grid size, square side length ($5\,\mathrm{mm}$), and a normal height distribution with a standard deviation of $1\,\mathrm{mm}$.

2) Video tracking and metrics: Locomotion trials are recorded using a high-speed camera (Nova S12, Photron). For each experimental setup, three trials were conducted on flat ground and steps, and five trials were conducted on rough terrain. For the flat ground and rough terrain trials, we track the front and rear hip joints from the side of the robot nearest the camera, and a point at the end of a front leg spoke in Tracker (Open Source Physics). In the Tracker App, a calibration stick sets the pixel-to-length scale, and tracking begins at the frame when the coils are actuated. Using the tracked points, the robot body's planar position (x, y) and velocity (v_x, v_y) were computed by averaging the positions and velocities of the front and rear hip markers (Fig. 1). To evaluate stability, the pitch angle was defined as the angle between the global x-axis and the line connecting the front and rear hip markers, with the front hip serving as the origin of the global coordinate frame. The RMS pitch amplitude for each trial was computed as the standard deviation of the pitch angle after the robot reached a stable locomotion speed.

C. Simulation and system identification

1) MuJoCo model with magnetic interactions: We simulate Milliquad locomotion using the MuJoCo rigid-body dynamics engine [18]. Although the parameter optimization method proposed in this work should be applicable to other simulators that support reasonable roll-out speeds or high parallelization, we chose MuJoCo given the tool's accessibility, fast computation, and high customizability. To simulate the magnetic torques used to actuate the robot, we added custom external torques based on (1). Each hip magnet is modeled as a point dipole $\mathbf{m_i} = m\mathbf{\hat{n}_i}$, where $\mathbf{\hat{n}_i}$ is the magnet axis in the ith hip frame and m is the magnetic moment. Using a variable that controls the desired field angle, (2), the corresponding torque on each joint is then calculated and applied every timestep.

At this size scale, the magnets at each joint have non-negligible effects on each other and nearby joints have additional torques due to the neighboring magnets' fields. To capture this effect, we also calculate the dipole-dipole interactions by computing the field at magnet i due to all other magnets using:

$$\mathbf{B_i} = \sum_{j \neq i} \frac{\mu_0}{4\pi |\mathbf{r_{j\to i}}|^3} \left[3(\mathbf{m_j} \cdot \mathbf{\hat{r}_{j\to i}})\mathbf{\hat{r}_{j\to i}} - \mathbf{m_j} \right] \tag{6}$$

where $\mathbf{r_{j\to i}} = \mathbf{x_i} - \mathbf{x_j}$ is the vector displacement from the ith magnet to the neighboring jth magnet. The internal torques are then calculated using $\tau_{\mathbf{int},\mathbf{i}} = \mathbf{m_i} \times \mathbf{B_i}$ and applied as a wrench on the corresponding magnet bodies.

Mechanical work from the external field is computed from simulated joint angular velocities $\omega_{\mathbf{j}}$ and external magnetic torques $\tau_{\mathbf{ext,j}}$ on each leg.

$$E_{total} = \sum_{t=0}^{T} \sum_{j}^{4} \tau_{\mathbf{ext,j}} \cdot \omega_{\mathbf{j}} \ \Delta t \tag{7}$$

Cost of transport is calculated as COT $= E_{total}/Mgd$, where M is the robot mass, d is distance traveled, and g is gravity.

The simulation captures the dominant physics of magnetic actuation and inter-magnet coupling, but omits several features of the physical robots. In particular, the clearance between the pin and hole in each hip joint introduces parasitic degrees of freedom, and preliminary experiments on isolated legs revealed highly nonlinear joint dynamics. However, we chose to simplify the simulation parameter space and instead represent joint losses with a single viscous damping coefficient, optimizable as a decision variable.

2) Parametric identification using a global optimizer: With a configurable simulation framework, we proceed to identify a set of contact and magnetic simulation parameters that can produce robot velocities that best match our experimental results. We fixed the simulation timestep to be $0.0005\,\mathrm{s}$ (for a $2\,\mathrm{kHz}$ simulation) to reduce the dimensionality of the optimization, and selected the fourth order Runge-Kutta integrator because MuJoCo's default semi-implicit Euler integrator produced unrealistic behaviors such as negative energy dissipation at this scale when paired with our custom actuation model.

Using velocity references (v^*) from all four robot morphologies across selected actuation frequencies and a parameter vector θ, we pose an optimization problem to solve for the best simulation parameters:

$$\theta^* = \arg\min_{\theta \in \Theta} J(\theta) \tag{8}$$

where the cost function $J(\theta)$ is a sum of costs across all test conditions (Φ) in the velocity references (v^*), as well as a regularization term which penalizes the variance in normalized velocity errors across trials, weighted by some factor λ:

$$J(\theta) = \sum_{\phi \in \Phi} C_\phi(\theta) + \lambda \mathbf{Var}_{\phi \in \Phi} \left(\frac{\bar{v}_\phi(\theta) - v_\phi^*}{v_\phi^*} \right) \tag{9}$$

The per-test-condition costs (C_ϕ) all contain metrics that can be extracted from a trajectory roll-out, including average simulated forward-velocity ($\bar{v}_\phi$), lateral displacement (d_ϕ^{lat}), final yaw (ψ_ϕ, used to determine if the robot ended up perpendicular to the field rotation), tumble amount (p_ϕ^{tumble}, used to measure if the robot undesirably completed a front-flip), and final progress defined as r_{prog}:

$$r_{prog} = \left(1 - \mathbf{clamp}\left(\frac{x_T - x_0}{x_{end} - x_0}, 0, 1 \right) \right). \tag{10}$$

This term is applicable only to the step terrain, comparing the robot's final forward position (x_T) to the end of the steps (x_{end}). The per-test-condition cost is a weighted sum of the aforementioned metrics (weighted by $w_{\_}$, summarized in Table I).

$$\begin{aligned} C_\phi(\theta) = & w_{vel} \left(\frac{\bar{v}_\phi(\theta) - v_\phi^*}{v_\phi^*} \right)^2 + w_{tum} \ p_\phi^{tumble}(\theta) \\ & + w_{lat} \ d_\phi^{lat}(\theta)^2 + w_{prog} \ r_{prog,\phi} \\ & + w_{yaw} \left[\max\left(0, \frac{\psi_\phi(\theta) - 60^\circ}{90^\circ}\right) \right]^2 \end{aligned} \tag{11}$$

Weights were varied across terrains to better capture experimental behaviors (e.g., simulated robots on rough terrain were more likely to tumble than their physical counterparts).

To evaluate the cost $J(\theta)$ for each parameter set, three trials were rolled out with randomized initial yaw sampled within $\pm 2^\circ$, and the median cost is chosen to represent that parameter's performance, except for when any initial condition leads to non-finite acceleration or simulation error, in which case that parameter set would get a failure cost of 1×10^6. Flat terrain simulation trajectories were trimmed to the average corresponding experimental trajectory time for the corresponding morphology and drive frequency. Step and rough terrain simulation trajectories were trimmed to the start and completion of the terrain.

This optimization was carried out using CMA-ES [21], with population size 16 and results converging before 4800 samples. Because we intend on using this simulation as a measurement tool, we subsample the population of final simulated trials with varied initial conditions to select the trials that best match the experimental metrics (3 closest trials for flat and step, 5 for rough terrain). As mentioned in the following section, we performed parameter optimization over individual terrains as opposed to pursuing a unified set of parameters to explain all experimental results.

III. Results

A. Manufacturing repeatability

We evaluated manufacturing repeatability by comparing two independently fabricated sets of robots with each morphology on flat ground at $10\,\mathrm{Hz}$ and $30\,\mathrm{Hz}$. For each morphology and frequency, we performed two trials and computed the mean forward velocity. The maximum percentage difference in mean velocity between two independently assembled robots with the same morphology was $8.8\,\%$ for L4 and $12.2\,\%$ for L1 at $10\,\mathrm{Hz}$ and $30\,\mathrm{Hz}$ respectively. This build-to-build variation is comparable to the trial-to-trial variability for a single robot under identical conditions (approximately $3.5\,\%$ for L4 and $25\,\%$ for L1), and smaller than the performance separation observed across morphologies in Fig. 3a. Therefore, we concluded that variation due to manufacturing does not introduce bias beyond the inherent stochasticity of locomotion at this scale or significantly affect the reported performance trends, and the remaining experiments were conducted using one set of robots.

B. Speed

We measured the forward progression speed (v_x^*) of the robots across the three terrains as the primary locomotion

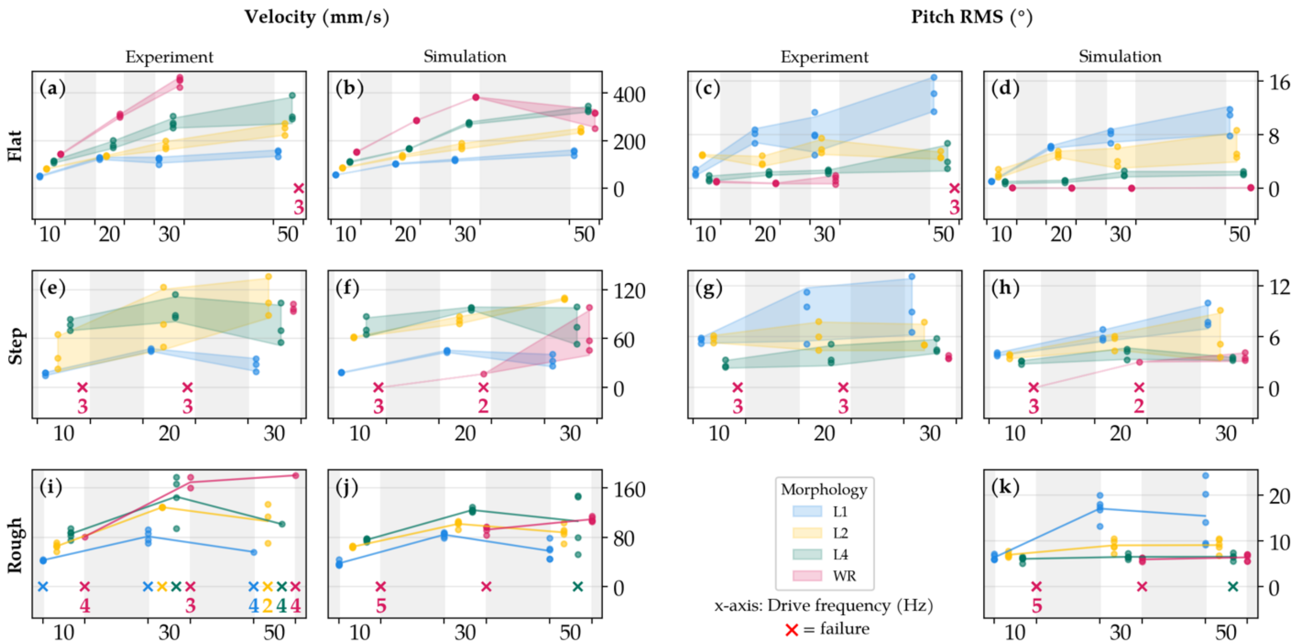

Fig. 3. Comprehensive comparison of experimental and simulated locomotion performance for four leg morphologies (L1, L2, L4, and WR) across flat, step, and rough terrains. The left two columns show forward velocity, and the right two columns show pitch RMS amplitude. For each metric, experimental results are presented alongside simulation results. Each individual marker denotes a measured data point from each trial, with shaded regions representing the standard deviation across trials. "×" markers indicate failure cases at the corresponding frequency, with the number of failed trials. Across different terrains, both morphology and actuation frequency influence locomotion speed and pitch dynamics. The rough terrain trials were recorded from an overhead view so the pitch angle of the robots could not be measured. The data points for each trial are staggered to improve visual clarity, and each unshaded band represents a discrete actuation frequency group.

performance metric. On flat ground, the WR morphology achieves the highest velocities from 10 Hz to 30 Hz, where it reached a maximum speed of $449.3\,\mathrm{mm\,s^{-1}}$ (or ≈ 50 body lengths per second) at 30 Hz (Fig. 3a). When actuated at 50 Hz, WR was not able to stably move forward because the wheels would lose sync with the magnetic field, causing chaotic wheel motion and yawing the robot perpendicular to the field. A modified startup strategy in which f_{drive} was ramped from 10 Hz to 50 Hz over 0.15 s did achieve a stable steady-state velocity of $709.4\,\mathrm{mm\,s^{-1}}$. However, because this protocol differs from the constant frequency tests, we exclude it from the comparison and instead use it as a qualitative description of achievable speeds on flat terrain. Across the legged morphologies, flat-ground velocity generally decreases with reduced spoke count, consistent with fewer effective ground contact events per cycle that can propel the body forward.

On steps (Fig. 3e), none of the morphologies were able to stably climb the steps at 50 Hz. The step terrain also highlights challenges for wheeled systems at this scale. The 1 mm step height is about 67 % of the robot's ground clearance (1.5 mm), and WR fails at 10 and 20 Hz due to insufficient momentum. This limitation is also commonly observed in larger-scale wheeled robots when encountering obstacles comparable to their wheel radius [22]. In contrast, L2 exhibits improved adaptability on this terrain, with an average forward speed on steps increasing linearly with drive

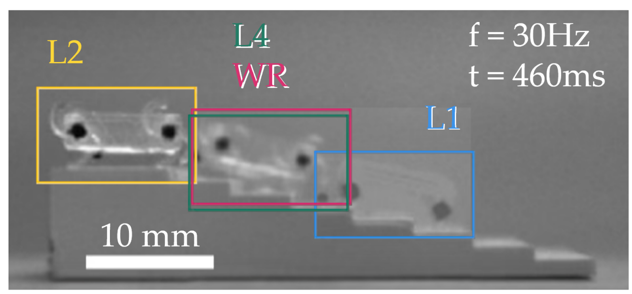

Fig. 4. Comparison of robot displacement along steps at 30 Hz. The frame corresponds to the moment when L2, the fastest configuration, reaches the end of the terrain at t = 460 ms.

frequency and reaching a maximum speed of $160.8\,\mathrm{mm\,s^{-1}}$ at 30 Hz. Fig. 4 shows a representative 30 Hz trial where L2 reaches the end of the steps at $t = 460\,\mathrm{ms}$ while the other morphologies lag behind.

On the rough terrain (Fig. 3i), L2 is the most robust morphology, achieving an average success rate of 80 % from 10 to 50 Hz. The highest average velocity reached $128.9\,\mathrm{mm\,s^{-1}}$ at 30 Hz, with the lowest standard deviation of $0.27\,\mathrm{mm\,s^{-1}}$. This indicates that L2 maintains a highly consistent speed when traversing rough terrain at 30 Hz, a level of consistency not achieved by the other configurations in any frequency. Fig. 5 shows the four robots' positions at the frame when the fastest robot (L4) reached the end of the

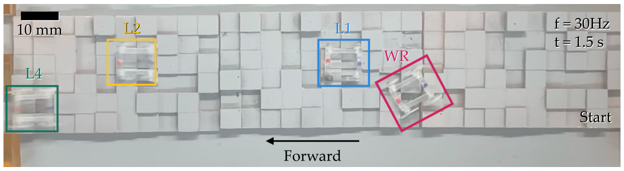

Fig. 5. Comparison of robot displacement on random rough terrain at 30 Hz. The frame corresponds to the moment when L4, the fastest configuration, reached the end of the terrain, whereas WR was immobilized at its current position and classified as a failure.

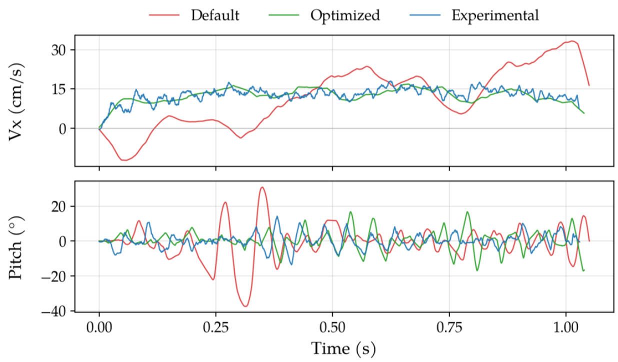

Fig. 6. Comparison between the simulated and experimental robot trajectory for L2 over flat terrain at 20 Hz. We include the trajectory with MuJoCo default parameters (red) as a comparison to the set of parameters optimized for flat terrain simulation (green).

terrain at 1.5 s (30 Hz drive frequency), while WR becomes stuck at the position shown and was classified as a failure.

C. Stability

To evaluate the stability of robot morphologies across terrains and actuation frequencies, we examined the RMS pitch amplitude of the robot base (Fig. 3c and Fig. 3g), which serves as a proxy for rotational disturbances that could affect onboard sensing and payloads. On both flat ground and steps, L1 exhibits increasing pitch RMS as drive frequency increases, indicating increased variability in the dynamic response of L1 at higher frequencies. In contrast, L4 and WR maintain consistently low pitch RMS on flat terrain (at most $5.5°$), consistent with frequent or near-continuous ground contact.

Experimentally, L2 exhibits an RMS pitch amplitude similar to that of L1 at low frequencies (10 Hz - 20 Hz), where both morphologies share a comparable walking-like pattern with periodic body contact. At 10 Hz, the body contact frequency is 10 Hz for L1 and 20 Hz for L2 as expected due to the two spokes. Above 30 Hz, L2 transitions away from body-ground contact and behaves more like a rolling morphology. Correspondingly, its pitch response becomes closer to that of L4 and WR. Pitch was not measured on rough terrain experimentally because those trials were recorded from an overhead viewpoint.

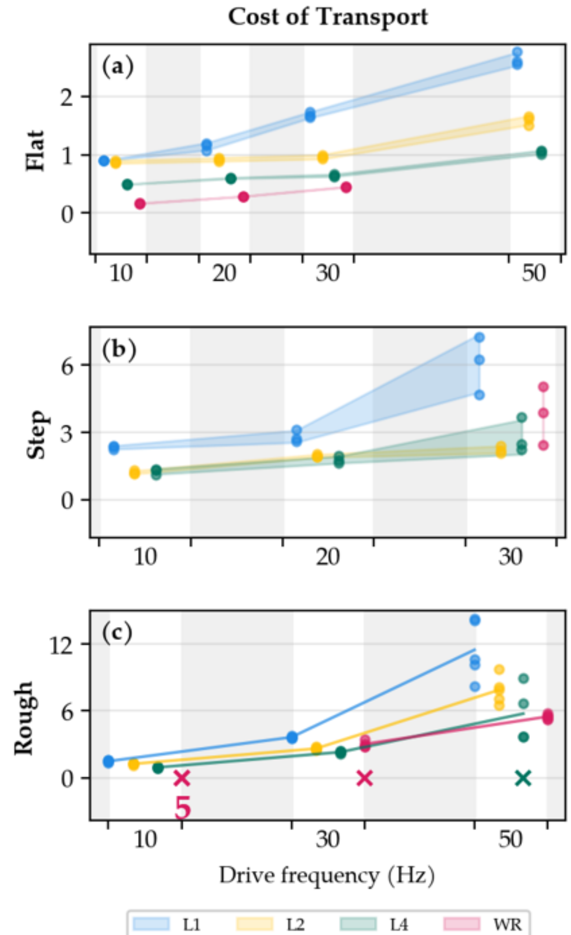

Fig. 7. The COT measured in simulation for four leg morphologies (L1, L2, L4, and WR) across flat, step, and rough terrains. Each individual marker denotes a measured data point from each trial, with shaded regions representing the standard deviation across trials. "×" markers indicate failure cases at the corresponding frequency, with the number of failed trials. The data points for each trial are staggered to improve visual clarity, and each shaded band represents a discrete actuation frequency group.

D. Real-to-sim parity

After performing parameter identification using CMA-ES to match simulated average locomotion velocities to experimental measurements for both flat and step terrains, we found that the optimization was unable to obtain a single parameter set that could reproduce behaviors across both scenarios. The discrete transition in WR's traversal of steps from failure at lower frequencies to success at 30 Hz was especially challenging to capture in simulation. We therefore proceeded to separate the velocity references into three batches and generated parameter sets θ_{flat}, θ_{steps}, and θ_{rough}, each of which provide consistent behavior within their respective terrains. The resulting sets of parameters (Table I) highlight that environment-specific contact properties can require substantially different values to match observed behavior, even if the materials remain consistent across terrains.

Fig. 6 illustrates the benefit of parameter optimization on

TABLE I

OPTIMIZED CONTACT PARAMETERS FOR EACH TERRAIN. RED HIGHLIGHTS OUTLIERS ($>1\sigma$ FROM MEAN ACROSS TERRAINS).

Parameter	Default	Flat	Step	Rough	Higher means:
`friction` (sliding)	1.0	0.493	0.734	0.640	more lateral sliding resistance
`friction` (torsional)	5.0×10^{-3}	6.11×10^{-4}	2.72×10^{-4}	2.35×10^{-5}	more spin resistance at contact
`friction` (rolling)	1.0×10^{-4}	1.11×10^{-6}	2.09×10^{-6}	1.48×10^{-6}	more rolling resistance at contact
`solref` (timeconst)	0.02	1.32×10^{-3}	2.42×10^{-3}	8.33×10^{-4}	softer, slower constraint stabilization
`solref` (dampratio)	1.0	2.74	1.53	3.67	more damped contact; <1 underdamped
`solimp` (dmin)	0.9	0.756	0.312	0.282	higher stiffness on initial touch
`solimp` (dmax)	0.95	0.927	0.850	0.508	higher stiffness under deep penetration
`solimp` (width)	1.0×10^{-3}	2.26×10^{-5}	1.53×10^{-5}	1.58×10^{-5}	more gradual stiffness ramp-up
`solimp` (midpoint)	0.5	0.697	0.925	0.814	full stiffness at shallower penetration
`solimp` (power)	2.0	4.93	4.85	5.43	more abrupt dmin→dmax transition
`moment_fudge`*	-	0.522	0.907	0.853	stronger magnet response to field
`field_fudge`*	-	1.49	0.802	1.50	stronger external drive field
`damping`	0	3.60×10^{-10}	2.98×10^{-10}	7.65×10^{-10}	more viscous joint resistance
`noslip_iterations`	0	1	30	0	more no-slip friction enforcement; 0 disables
`noslip_tolerance`	1.0×10^{-6}	7.57×10^{-6}	1.99×10^{-5}	1.01×10^{-6}	more residual slip allowed
`o_margin`	0	3.03×10^{-4}	5.74×10^{-4}	3.00×10^{-5}	larger contact detection distance
w_{vel}		5	5	5	weight on velocity error in Eq. (11)
w_{tum}		1	1	2	weight on tumble penalty
w_{lat}		5	1	5	weight on lateral deviation
w_{prog}		0	2	0	weight on velocity variance
w_{yaw}		1	0	1	weight on yaw penalty

*Custom scaling factors, not MuJoCo parameters.

a representative L2 trial at 20 Hz on flat terrain. With default MuJoCo parameters, motion is chaotic, inconsistent with the robot behavior in experiments. After optimization, the simulated forward velocity and pitch are far better matched to the experiment, illustrating both the difficulty of simulating locomotion at this scale and the substantial improvement achieved through system identification. Given the selected simulation data presented in Fig. 3, the optimized simulations achieve mean velocity error of 6.1 % on flat terrain and 11.8 % on steps. Most importantly for our goal of using this simulation as a measurement tool, the overall trends of speed and pitch between morphologies in Fig. 3 are captured.

In addition, the simulation reproduces the qualitative failure trend that the WR morphology is comparatively less reliable across non-flat terrains, but also tends to produce pitch predictions with smaller variance than observed experimentally. The largest discrepancies in speed occur at higher frequencies and on rough terrain, where intermittent contacts are more likely to expose limitations in the simulator's contact model.

E. Cost of transport

Directly measuring the mechanical cost of transport in the physical system is challenging under external magnetic actuation. We can calculate an upper bound on mechanical cost of transport by substituting the maximum possible torque on each hip magnet ($\tau_{max} = |\mathbf{B}_{\mathbf{ext}}||\mathbf{m}|$) in Eq. 7 as a good sanity check, but this will not provide insights into how morphologies might differ. We therefore use the calibrated simulation as a measurement tool to compute the mechanical cost of transport (COT) from the external work done on each joint and the distance traveled. We are primarily interested in trends related to each morphology, terrain, and drive frequency as summarized in Fig. 7.

Across terrains, COT generally increases with actuation frequency and speed, consistent with higher mechanical power demands at higher drive frequencies. On flat terrain, WR appears to maintain a fairly stable COT across frequencies, with reduced spoke count trending toward higher COT. This is not surprising given that the robots tend to undergo more height variation over each stride at lower spoke counts, whereas the height of the WR morphology stays steady. On steps, L2 provides a favorable speed-efficiency tradeoff relative to the other morphologies, which makes sense given its higher step-climbing speeds in experiments. On rough terrain, COT differences between morphologies are less pronounced than on flat terrain and steps, reflecting the increased influence of intermittent contacts and disturbances during traversal.

IV. CONCLUSIONS AND FUTURE WORK

This work addresses two coupled challenges in the design of tiny robots: selecting an effective locomotion morphology and obtaining the quantitative data needed to assess that choice. Selecting a morphology for small-scale robotic locomotion is often challenging due to a lack of prior knowledge about the terrain a robot may interact with, even with relatively clear task specifications. For terrain adaptability, L2 demonstrates superior success rates (80 %) over rough terrain trials while maintaining reasonable pitch and speed over the flat and step terrains. We hypothesize that L2's success emerges from its ability to contact the ground in a leg-like manner at lower frequencies, while approximating rolling contact at higher frequencies, effectively accessing both locomotion regimes. Although L4 marginally improves upon L2's performance in speed and efficiency, its lackluster performance on rough terrain, much like that of WR, indicates that it may not be the best choice for small robots when terrain roughness can be similar to robot wheel radius. If a task specifies locomotion over flat terrain and speed is a factor, then WR is best given its highest speed and lowest COT. Contrary to concerns about rotary joints at small scales, pin joints and wheels at this scale functioned reliably, suggesting that these challenges may be less severe at the $100\,\mathrm{mg}$ scale.

Beyond the morphology comparison, this work demonstrates the utility of physics-based simulation combined with system identification as a design and analysis tool for small-scale robots. By fitting uncertain contact and magnetic parameters to experimental results, we obtain MuJoCo models that qualitatively reproduce locomotion behaviors, including terrain-dependent failure modes, and provide estimates of mechanical energy expenditure that are otherwise inaccessible at this scale. In the future, the simulation could also be used to extract information on required torque (useful for designing actuators on more autonomous systems) and diagnostic information regarding environmental interactions like ground reaction forces that can be used to improve future robot designs.

A primary limitation of this work lies in the requirement for terrain-specific parameter sets; a single set of simulation parameters was insufficient to describe all experimental data. As such, the current approach is limited in its use as a predictive design tool, but remains valuable for extracting otherwise inaccessible quantities like COT from existing experimental data. One possible approach to improve the simulation is to increase the complexity of the robot model to include improved approximations for joint slop and non-linear damping by increasing the joint degrees of freedom and adding custom forces, respectively. Such attempts can improve the representational capacity of the model, with the consequence of increasing the search space that the optimizer has to explore. We were also only able to experiment with a limited number of terrain parameters, such as a fixed step width, step height, random terrain height standard deviation, or even contact friction. Future work can perform a more comprehensive comparison over ranges of terrain or robots with even wider design spaces, such as limb stiffness, number of appendages, or number of body segments toward an improved understanding of tiny robot design for effective and efficient locomotion.

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